Zachary A. Bolton.

Staff ML Engineer · AI Evaluation & Reliability

U.S. citizen · Based in Brazil · Remote preferred · Open to relocation

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> Staff ML engineer building real-time AI systems and the evaluations that govern their release. I've led engineering teams, trained speech models, and made eval-driven development a shared practice. Now bringing that experience into AI safety: behavioral evaluations, adversarial testing, and research infrastructure.

Experience

Boostlingo Staff Machine Learning Engineer

August 2025–present · Promoted to Staff September 2026

- Lead engineer for a real-time AI interpreter; led four engineers. Built C#/.NET and Orleans voice infrastructure spanning voice-activity detection, conversation management, telephony, and rollout controls.
- Drove the organization's shift to eval-driven development through shared tooling, tech talks, and hands-on coaching. Built multilingual datasets, automated scoring, provenance records, and required CI gates; helped teams beyond the interpreter adopt the practice.
- Evaluated 29 languages and worked with language experts on release criteria, informing the rollout to 15 generally available languages by October 2026. Added independent language-ID checks after quality scores missed wrong-language output.
- Developed a modular intervention framework and a direct-address guard to keep the interpreter in role. Built spoken adversarial tests with ordinary-speech controls and detector-on/off comparisons.
- Fine-tuned a pyannote speech model on ~92 hours of staged conversations for crosstalk detection. Fixed evaluation and codec defects, tested on held-out meeting audio, and shipped detection in shadow mode, with live interruptions disabled.
- Designed, conducted, and authored a public medical AI interpreting accuracy study. Lead presenter for MT Evaluation for Spoken AI Interpreting at AMTA 2026.

Kid Chess Software Engineer & Director of Technology

2015–August 2025

- Moved 4,000 students online during the pandemic, helping retain over 90% of staff and keep the company afloat. Enabled a full return to in-person teaching plus a continuing online revenue stream.
- Grew engineering and QA from two to nine; hired, trained, and led delivery. Owned the online platform's product, engineering, and operations.
- Built LLM student-feedback reports with an anti-sycophancy stage and rubric-based human review to catch unsupported praise and guide improvements.

Research & safety experiments

University of London Independent ML Research

April 2024–April 2025

- Built a Siamese SBERT authorship-verification pipeline in Python/PyTorch on AWS: genre-balanced data, text-distortion experiments, and external VALLA benchmarking. Earlier work explored stylometry and neural models.
- Investigated apparently perfect Brier scores: logged predictions, reran experiments, and traced the result to a scoring artefact. Documented methods and limitations in a public research report.
- **Independent safety experiments, AI-assisted:** built controlled résumé prompt-injection tests in Inspect, with visually identical PDFs, a clean control, and decision-based scoring. In a Petri behavioral-auditing work sample, separated attempted from completed actions and identified judge inconsistencies and simulator drift.
- **Harmful-request classification:** compared TF-IDF and DeBERTa, documenting false positives, generalization limits, and reproducibility. AI-assisted implementation and analysis.
- **Gender bias in chess:** analyzed discussions before and after *The Queen's Gambit*; documented sampling limits and how positive sentiment scores can miss sexist content.

Education & tools

BSc Computer Science, Machine Learning & AI — University of London, First Class, 2025 · 90% average.

AGI Strategy — BlueDot Impact — Certificate earned November 10, 2026 · 30-hour program.

Python · PyTorch · Hugging Face · Inspect AI · C#/.NET · Orleans · Datadog LLM Observability · AWS · Git/CI

AI safety engagement

Self-directed study since 2016, beginning with Robert Miles. Developing an eval proposal from AGI Strategy on whether AI notetakers favor the more powerful or fluent speaker. Intend to take Technical AI Safety next. DeepLearning.AI, *Red Teaming LLM Applications*, January 2026.